

Problem Statement & Proposed Solution

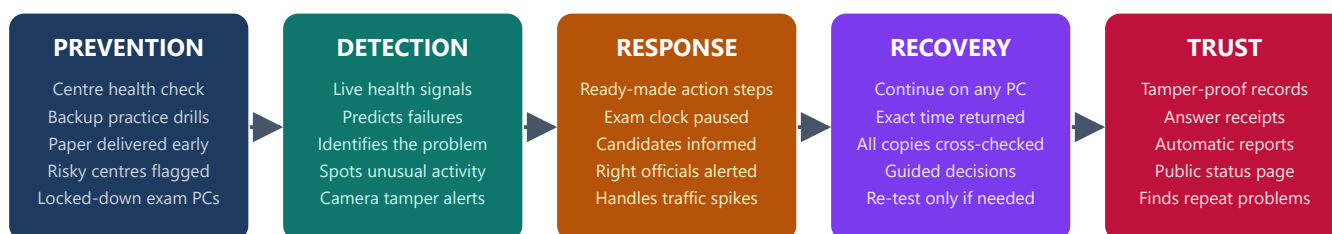
Resilient & Trustworthy Online Assessment Ecosystem · ASTRA

Part A: The Problem

A1. The challenge in brief

Most large exams in India are now computer-based tests (CBT). When something goes wrong (a technical fault, poor planning, a security breach, or mismatched data), exams get delayed or have to be held again. This costs money and causes great hardship for candidates, and it makes students lose trust in the exam system.

The challenge is to build an exam system that is **reliable, transparent, secure and able to recover quickly** when something unexpected happens, at every stage of the exam:



A2. What the problem statement asks for: 12 abilities

#	Ability	What it means in simple terms
1	Real-time monitoring	See the condition of every centre and every candidate's exam, live
2	Early detection and prediction	Warn before something breaks
3	Automatic incident handling	Work out what went wrong (one PC, a room, a server, a vendor) and alert the right person
4	Backup and recovery	No answer is lost; no candidate starts again from zero
5	Tamper-proof data	Any change to answers or records can be detected, even changes made by staff
6	Spotting unusual patterns	Flag suspicious activity for a human to check
7	Checking all answer copies	After a problem, confirm every saved copy matches
8	Informing candidates	Tell candidates what's happening within minutes, not hours
9	Decision help	Help officials decide quickly and consistently whether to reschedule or re-conduct
10	Fairness	Candidates hit by the same problem are treated exactly the same
11	Records and reports	Everything that happened can be reconstructed afterwards
12	AI analysis of repeat problems	Find and fix problems that keep recurring

Expected result: exams that are more reliable, secure, transparent and trustworthy, and that **avoid re-conducting exams unnecessarily**.

A3. Evidence: digital and technical failures only

18

digital failures studied,
2009–2026

489

centres lost one CUET-UG
shift

2,445

NEET-PG candidates at one
failed centre

3

MP failures: security,
postponement, server crash

Exam delivery failures: power, servers, vendors

When	Exam	What went wrong
2009	CAT	Computer viruses on centre PCs; about 7–8 thousand of 70 thousand candidates affected
2022	JEE Main	Questions didn't load and servers went down; the Supreme Court ordered an extra exam session
2022	CUET-UG	Question paper uploaded at 5 pm for a 3 pm exam; all 489 centres lost the shift
2025	JEE Main	Technical fault at a centre; 114 candidates had to come back another day
2025	BITSAT	Server failures and long waits in several cities
2025	CUET-UG	Technical problems across India; late entry and server errors
2025	SSC Phase 13 and CGL	After a new vendor took over: frozen computers, broken keyboards and mice, server crashes; about 28 lakh CGL candidates
2026	CUET-UG	A fault at the technology partner's end delayed the start by over 2 hours
2026	NEET-PG and AIIMS exam, Jaipur	Repeated power failures at a computer-test centre; the same centre failed two days in a row ; logs were asked for only afterwards
2025 MP	MPESB Group-1	Shift postponed after 4,818 candidates had already reached 42 centres
2026 MP	MPESB Forest Guard / Jail Prahari	The exam company's server failed; candidates waited about 3 hours ; shift cancelled

Digital security failures

When	Exam	What went wrong
2020–21 MP	3 MPESB recruitment exams	A state technical inquiry (by MPSEDC) found question papers were downloaded from unauthorised computers ; all 3 exams cancelled
2018	SSC CGL	Remote-control software on exam PCs let outsiders answer for candidates
2021	JEE Main	Same method: exam PCs at a centre were controlled remotely
2025	SSC / JSSC CGL, Dhanbad	The centre's own server was tampered with; answers were selected automatically

Digital marking and result failures

When	System	What went wrong
2017	Mumbai University online marking	Results of 477 exams delayed by 3–4 months

When	System	What went wrong
2019	Telangana Class 11–12 results	The vendor's software produced wrong mark sheets
2026	CBSE online marking	Wrong or missing answer-sheet scans, overloaded website, exposed passwords

A4. Why these failures happen

**No live view**
Officials find out late (Jaipur, MPESB 2026)

**No automatic backup**
One fault stops the exam (SSC 2025)

**One weak link stops everything**
A single paper-delivery system or vendor server (CUET 2022, MPESB 2026)

**No checks before the exam**
Power backup and computers aren't tested beforehand

**Insiders not watched**
Access to computers and servers misused (MPESB 2021, Dhanbad)

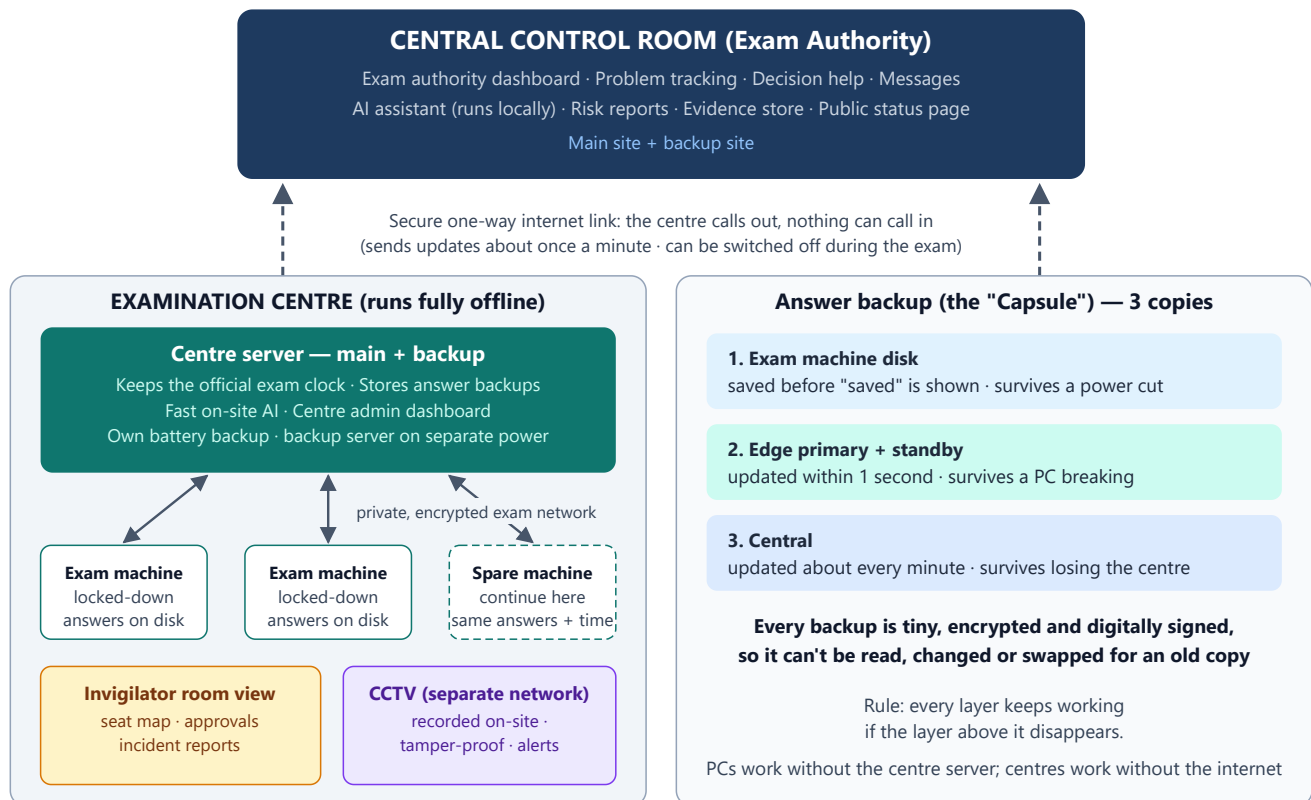
**Poor communication**
Hours of silence; no clear rules for giving back lost time

Part B: The Solution: ASTRA

Failures pause. Exams don't.

Each part keeps working even if the part above it fails: exam PCs keep working if the centre server fails, and centres keep working if the internet fails.

B1. How it is built



- **Exam PCs** save each candidate's answers on the spot.
- **The centre server** is a small local server inside each centre, with a spare. It runs the exam, keeps the official clock and holds a copy of every answer.
- **The central control room** oversees all centres, helps with decisions and keeps the evidence.

B2. The core idea: the answer backup ("Capsule")

1–3 KB

per candidate: smaller than a single photo

< 1 sec

to copy each answer to the centre's servers

3 copies

exam PC · centre server · control room

0 min

of exam time lost during an outage

- Each answer is **saved to the PC's disk before "saved" appears on screen**, so a sudden power cut can't erase it.
- The **centre server keeps the official exam clock**, which stops while the exam is paused. Candidates get back every minute lost.
- After a failure, the invigilator approves and the candidate **continues on any PC**, with the same questions, same answers and exact time left.

- The backup is **encrypted** (unreadable without a key) and **digitally signed** (any change is detected). **Only one PC can use a candidate's exam at a time.**

B3. All 12 abilities covered

#	How ASTRA does it	Where AI helps
1	Live dashboards for invigilators, centres and the exam authority; out-of-date data turns grey	Spots centres behaving unusually
2	Health check and backup drill at every centre before the exam	Predicts failures; ranks risky centres
3	Rules that sort problems by type and seriousness	Classifies incidents; reads Hindi/English reports from invigilators
4	Answer backup, spare server, continue on any PC or even at another centre	–
5	Encryption, digital signatures, tamper-proof records, answer receipts	–
6	Automatic consistency checks	Flags oddities, e.g. answers entered with no keyboard or mouse use
7	Compares all three copies after any problem	Flags mismatches
8	Messages within 30 seconds, public status page, notice board at the gate	Drafts messages in Hindi and English
9	5-step decision guide; two officials must approve	Estimates repair time; plans moving candidates; explains its advice
10	Exact record of each candidate's active exam time	–
11	Tamper-proof log of every event; automatic incident reports	Writes report summaries
12	Reports by centre, vendor, region and cause	Finds repeat problems and their causes

B4. The 5-step decision guide: avoiding needless re-exams

Step	Action	When
0	Continue; give back lost time	Short outage, answers safe
1	Extend the exam for that room	Longer outage
2	Move affected candidates to another room or centre the same day	Centre can't be used
3	Re-test only the affected candidates later	Very long outage
4	Hold the whole exam again	Widespread damage or a security breach

The system shows its reasons; officials decide. Steps 3 and 4 need two senior officials to approve.

B5. The same incidents, with ASTRA

What happened

- **Jaipur 2026:** same centre failed twice; re-tests a week later
- **MPESB 2026** : server down, 3-hour wait, shift cancelled
- **CUET 2022:** paper arrived late, 489 centres cancelled
- **MPESB 2021** : papers copied from unauthorised computers

With ASTRA

- Day-1 power blips mark the centre as risky, so the next exam is moved; any outage becomes a pause
- Centres don't depend on the company's server; candidates informed in under 30 seconds
- The locked paper reaches centres hours early; only a small unlock key is needed at start time
- Paper stays locked until exam time; every access needs two people and is recorded

B6. Monitoring, CCTV and security

Who sees what

Invigilator: their room. Centre head: their centre. Exam authority: every centre, down to one seat. **Nobody can see candidates' answers during the exam.**

CCTV

Recorded inside the centre on a separate network. Every clip is fingerprinted so edits show. Alerts if a camera is covered, moved, switched off or looping old footage.

★ The centre server as a "blind courier"

Like a courier carrying a sealed envelope: the centre server stores and passes on answers but **can't open or change them**. This stops the kind of insider tampering seen in Dhanbad.

★ Sabotage becomes pointless

If cutting the power only pauses the exam, nobody gains anything by doing it. We tested the design against about 80 ways of attacking it.

Sources

CAT 2009 · JEE Main 2022 · CUET-UG 2022 · JEE Main 2025 · BITSAT 2025 · CUET-UG 2025 · SSC Phase 13 · SSC CGL 2025 · CUET-UG 2026 · Jaipur 2026 · NEET-PG Jaipur · MPESB 2025 · MPESB 2026 · MPESB 2020–21 · SSC 2018 · JEE Main 2021 · Dhanbad 2025 · Mumbai Univ. 2017 · Telangana 2019 · CBSE 2026